

TEAM CONTRIBUTION REPORT

1. Introduction

The **Employee Attrition Prediction System** was successfully developed through the collaborative efforts of **Dimpal Sharma** and **Vaibhav Magar**. The project followed a structured software development and machine learning lifecycle consisting of requirement analysis, dataset collection, preprocessing, exploratory data analysis, machine learning model development, performance evaluation, documentation, and presentation.

To ensure efficient project execution, responsibilities were distributed based on different phases of the project while maintaining continuous collaboration throughout development. Regular discussions, testing sessions, and documentation reviews ensured consistency and quality across all project components.

2. Project Development Phases

Phase	Description
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Phase 1	Problem Identification & Requirement Analysis
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Phase 2	Dataset Collection
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Phase 3	Data Preprocessing
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Phase 4	Exploratory Data Analysis
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Phase 5	Feature Engineering & Scaling
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Phase 6	Machine Learning Model Development
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Phase 7	Model Evaluation
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Phase 8	Result Analysis
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Phase 9	Documentation
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Phase 10	GitHub Repository & Presentation
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3. Individual Responsibilities

Team Member 1: Dimpal Sharma

Primary Responsibilities

Requirement Analysis

- Identified the project problem statement.
- Selected Employee Attrition Prediction as the project topic.
- Designed the complete project workflow.

Dataset Collection

- Selected IBM HR Analytics Employee Attrition Dataset from Kaggle.
- Verified dataset quality.
- Understood dataset attributes.
- Studied feature descriptions.

Data Preprocessing

- Missing value analysis.
- Duplicate record checking.
- Label Encoding.
- Feature Scaling.
- Dataset cleaning.

Exploratory Data Analysis

Generated multiple visualizations:

- Correlation Heatmap
- Count Plot
- Histogram
- Box Plot
- Pair Plot

- Pie Chart
- Monthly Income Distribution
- Gender Analysis
- Job Role Analysis

Machine Learning Development

Implemented

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- KNN

Performed

- Model Training
- Model Testing
- Accuracy Calculation
- Performance Comparison

Team Member 2: Vaibhav Magar

Primary Responsibilities

Result Analysis

- Compared machine learning models.
- Analysed prediction accuracy.
- Evaluated Precision.
- Evaluated Recall.
- Evaluated F1 Score.
- Prepared model comparison.

Documentation

Prepared

- Project Report
- README
- References
- Bibliography
- Result Discussion
- Conclusion

GitHub Repository

Managed

- Repository Creation
- Folder Structure
- Version Control
- Source Code Upload
- Dataset Upload
- Screenshots
- Report Upload

Presentation

Prepared

- PowerPoint
- Workflow Diagram
- Output Screenshots
- Demonstration Slides

Testing

Performed

- Output Verification

- Bug Testing
- Code Validation
- Final Testing

4. Collaborative Contributions

Although responsibilities were distributed, both members actively participated in every important stage of the project.

Both members jointly contributed to:

- Project Planning
- Dataset Understanding
- Brainstorming
- Feature Selection
- Model Comparison
- Code Review
- Documentation Review
- Final Demonstration
- Presentation Preparation

5. Contribution Matrix

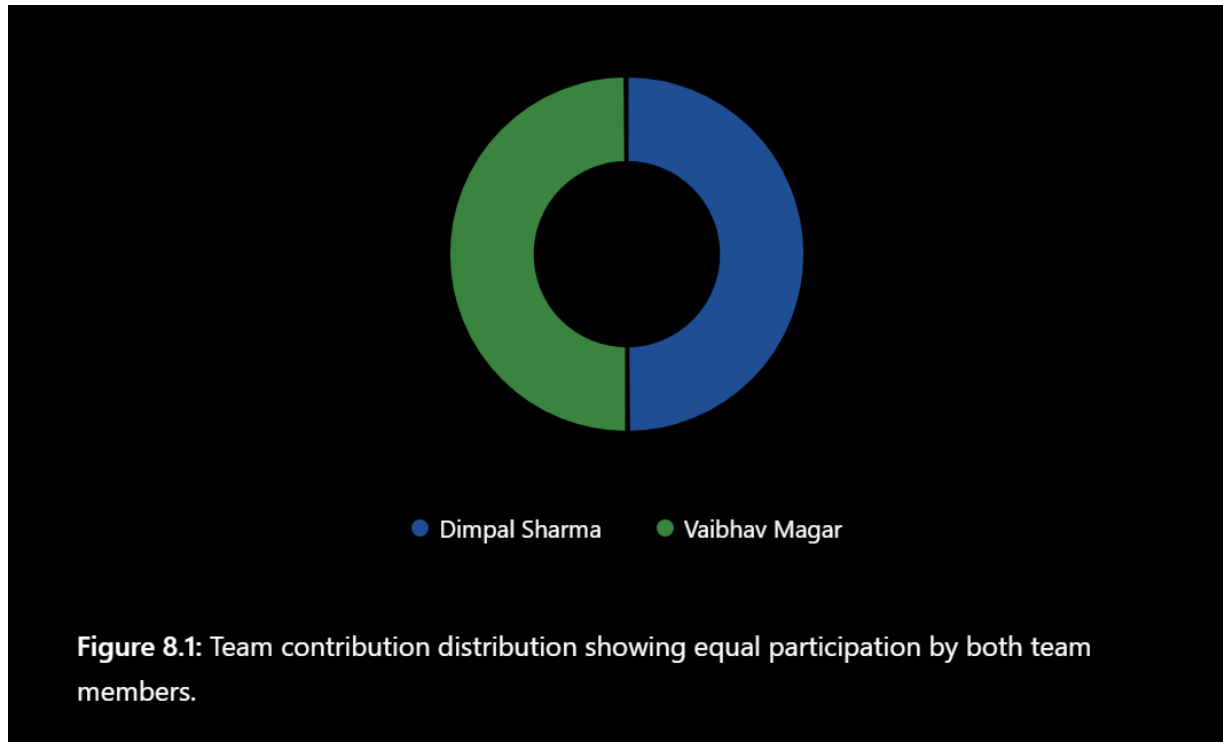
Activity	Dimpal Sharma	Vaibhav Magar
Requirement Analysis	✓	✓
Dataset Collection	✓	✓
Dataset Cleaning	✓	✓
Missing Value Handling	✓	✓

Activity	Dimpal Sharma	Vaibhav Magar
Duplicate Removal	✓	✓
Label Encoding	✓	✓
Feature Scaling	✓	✓
EDA	✓	✓
Data Visualization	✓	✓
Logistic Regression	✓	✓
Decision Tree	✓	✓
Random Forest	✓	✓
SVM	✓	✓
KNN	✓	✓
Model Evaluation	✓	✓
Accuracy Comparison	✓	✓
Result Analysis	✓	✓
GitHub Repository	✓	✓
Report Writing	✓	✓
PPT Preparation	✓	✓
Final Testing	✓	✓

6. Contribution Distribution

Team Contribution Distribution

Percentage contribution of each team member.



7. Conclusion

The successful completion of the **Employee Attrition Prediction System** was made possible through effective teamwork and continuous collaboration between **Dimpal Sharma** and **Vaibhav Magar**. Both members actively participated in every stage of the project, including planning, implementation, machine learning model development, testing, documentation, GitHub repository management, and presentation. The combined efforts resulted in a well-documented and fully functional machine learning project that satisfies the academic requirements and demonstrates practical application of data science techniques.